

ELEC 305 Project 1

Dohoon Kwag, Anthony Zhang

April 2026

1 Low Power 5T OTA

This design utilizes a 5T OTA topology to satisfy a power consumption lower than $1\mu W$. By selecting a channel length of $2\mu m$, the design minimizes channel-length modulation, which provides sufficient output resistance to achieve a DC gain exceeding 42 dB. The input differential pair was sized with a high g_m/I_D ratio to maximize transconductance efficiency, ensuring the 600 kHz GBW target was met with minimal current draw.

Table 1: Low-Power 5T OTA — Simulated Performance vs. Target Specifications

Parameter	Specification	Simulated
DC Gain	≥ 42 dB	42.34 dB
Gain-Bandwidth Product (GBW)	≥ 600 kHz	734.66 kHz
Phase Margin	$\geq 60^\circ$	88.18°
DC Power Consumption	$\leq 1 \mu W$	864 nW
Supply Voltage V_{DD}	≤ 1.8 V	1.8 V
Load Capacitance C_L	1 pF	1 pF
Reference Current I_{bias}	100 nA	100 nA
Temperature	27 °C	27 °C
Process Corner	TT	TT

1.1 Theoretical Analysis

We can determine the small-signal parameters using the g_m/I_D methodology centered on the $1\mu W$ power constraint. The drain current I_D flowing through each branch is determined by the tail current source $M5$, which is mirrored from $I_{bias} = 100$ nA. Based on the power budget of $P \leq 1\mu W$, a mirror ratio of $k = 4.55$ was chosen ($W_5/W_6 = 4.55$):

$$I_{TAIL} = 4.55 \cdot I_{bias} = 4.55 \cdot 100, \text{ nA} = 455, \text{ nA}. \quad (1)$$

This tail current splits evenly between the two differential branches, resulting in a quiescent branch current of:

$$I_{D1,2} = I_{D3,4} = \frac{I_{TAIL}}{2} = 227.5, \text{ nA}. \quad (2)$$

The GBW is determined by the transconductance (g_m) of the input differential pair and the load capacitance (C_L):

$$GBW = \frac{g_m}{2\pi C_L} \quad (3)$$

Given that $(GBW) \geq 600$ kHz and $C_L = 1$ pF, the transconductance g_m must be at least $3.77 \mu\text{S}$.

The DC gain is determined by the transconductance and the parallel combination of the output resistances of $M2$ and $M4$. From the $L = 4 \mu\text{m}$ LUT at the operating point, we find the intrinsic gain (g_m/g_{ds}):

$$\left(\frac{g_m}{g_{ds}}\right)_N = 267.5 \quad (4)$$

$$r_{o,N} = \frac{267.5}{3.77 \times 10^{-6}} = 70.95, \text{ M}\Omega. \quad (5)$$

Assuming similar intrinsic gain for the PMOS load ($r_{o,P} \approx r_{o,N}$), the total output resistance R_{out} and DC gain A_v are:

$$R_{out} = r_{o,N} \parallel r_{o,P} \approx 35.47, \text{ M}\Omega. \quad (6)$$

$$A_v = g_{m1} \cdot R_{out} = (3.77 \times 10^{-6}) \cdot (35.47 \times 10^6) = 133.75, \text{ V/V}. \quad (7)$$

In decibels, this provides:

$$A_v(\text{dB}) = 20 \log_{10}(133.75) = 42.52, \text{ dB}, \quad (8)$$

satisfying the $A_v \geq 42$ dB specification.

The width of the transistors is determined by applying the current density (I_D/W) obtained from the LUTs. For the selected channel length of $4 \mu\text{m}$ and the operating efficiency of 16.55 V^{-1} ($3.277 \mu\text{S}/227.75 \text{ nA} = 16.55 \text{ V}^{-1}$), the LUT specifies a current density of $0.307 \mu\text{A}/\mu\text{m}$. To achieve the required branch current of 227.5 nA , the width (W) for the input pair transistors ($M1$ and $M2$) is calculated as:

$$W_{1,2} = \frac{I_D}{(I_D/W)_{LUT}} = \frac{227.5 \text{ nA}}{307 \text{ nA}/\mu\text{m}} \approx 0.74 \mu\text{m} \quad (9)$$

Assuming a dominant-pole response where non-dominant poles are located at significantly higher frequencies due to the low-power nature of the circuit, the phase margin is approximated by the single-pole lag at the unity-gain frequency:

$$\text{PM} \approx 180^\circ - 90^\circ = 90^\circ. \quad (10)$$

1.2 Calculated vs Simulated Performance

The theoretically derived parameters were implemented within Xschem using the Skywater 130nm PDK. The schematic made in Xschem and the simulation results are shown below.

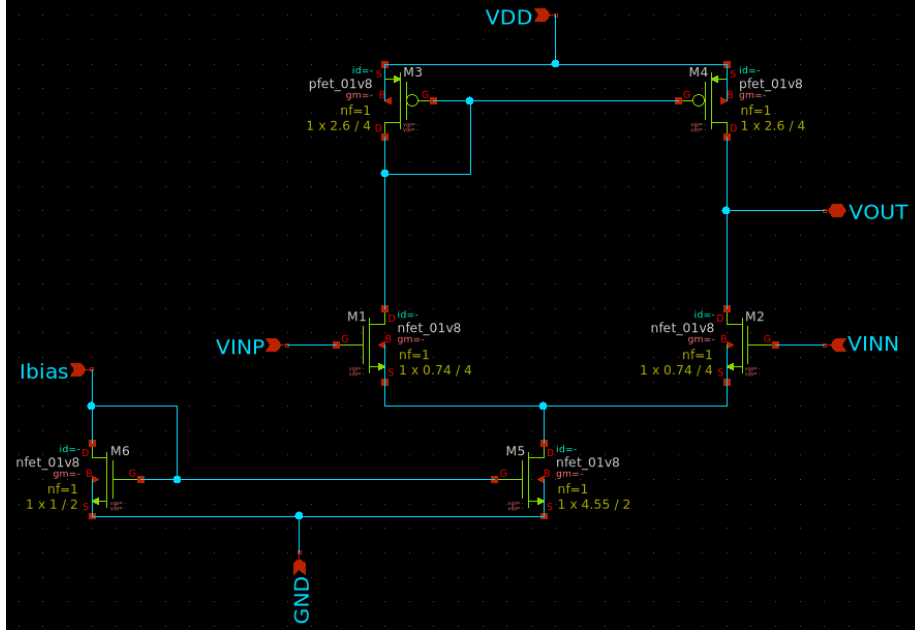


Figure 1: Complete Schematic of Low Power 5T OTA

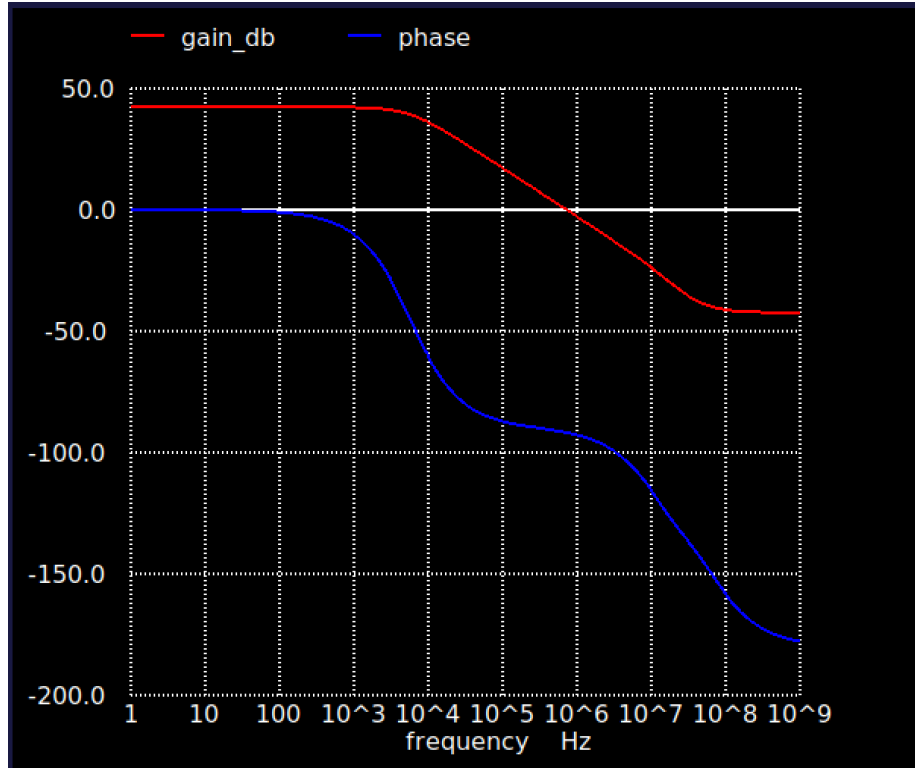


Figure 2: Simulation results of Low Power 5T OTA

```

No. of Data Rows : 901
binary raw file "/headless/.xschem/simulations/5t_ota_ac.raw"
dc_gain          = 4.234235e+01
fu               = 7.346609e+05
phase            = -9.182458e+01
dc_gain = 4.234235e+01
fu = 7.346609e+05
phase = -9.18246e+01
ngspice 1 -> █

```

Figure 3: Simulation results of Low Power 5T OTA

The resulting output confirms that the design targets were successfully achieved. The simulation recorded a DC gain of 42.342 dB and a unity-gain frequency (f_u) of 734.66 kHz, which was very similar to our theoretical DC gain of 42.52 dB and GBW of 600 kHz. The larger GBW can be attributed to the fact that the LUDs were generated at a DC operating point of $V_{DS} = 0.6$ V; however,

in the simulation, the NMOS input transistors do not operate at this V_{DS} . As such, g_{m1} is higher than estimated, causing the GBW to be larger in the simulation. In general though, these values exceed the minimum specifications of our minimum specs required for the low power 5T OTA, proving this design procedure works.

The phase at the unity-gain frequency was measured at -91.82° . This result indicates a Phase Margin (PM) of approximately 88.2° ($180^\circ - 91.8^\circ$), which is well above the stability requirement of 60° . This phase margin is also very close to our predicted 90 degree phase margin. The difference can be attributed to the fact that our predicted calculated neglects the non-dominant pole at the gate of the PMOS current mirror, located at:

$$f_{p2} \approx \frac{g_{m3}}{2\pi \cdot 2C_{gs3}}. \quad (11)$$

This pole introduces additional phase lag at frequencies approaching f_u , which then explains why the simulated phase margin is lower than the estimated phase margin. Even though the simulated PM is below the hand-calculated value, the simulated PM still exceeds the 60° specification with a large margin.

2 High Speed 5T OTA

This design utilizes a 5T OTA topology in order to achieve a gain-bandwidth (GBW) product of at least 200 MHz and DC gain of at least 37 dB. In order to satisfy this, we choose a short channel length of 1000 nm for all transistors in order to minimize time taken for charge carriers to move from source to drain, thus increasing GBW. To achieve the balance between gain and bandwidth, we choose a lower g_m/I_d of around 7, which also would allow our power consumption to be lower than the requirement of $800 \mu W$.

Table 2: High-Speed 5T OTA — Simulated Performance vs. Target Specifications

Parameter	Specification	Simulated
DC Gain	≥ 37 dB	40.50 dB
Gain-Bandwidth Product (GBW)	≥ 200 MHz	226.85 MHz
Phase Margin	$\geq 60^\circ$	72.625°
DC Power Consumption	$\leq 800 \mu W$	$540 \mu W$
Supply Voltage V_{DD}	≤ 1.8 V	1.8 V
Load Capacitance C_L	0.5 pF	0.5 pF
Reference Current I_{bias}	100 μA	100 μA
Temperature	$27^\circ C$	$27^\circ C$
Process Corner	TT	TT

2.1 Theoretical Analysis

In order to calculate the theoretical gain, GBW, and PM of the 5T OTA, we first determined the small-signal transconductance (g_m) of the NMOS input pair and the PMOS current-mirror load.

The drain current I_D flowing through each branch of the differential pair is determined by the tail current source, which is mirrored from the reference current I_{bias} . Since $W_5/W_6 = 40/20 = 2$ and $L_5 = L_6 = 1 \mu\text{m}$, the mirror ratio is exactly 2, giving:

$$I_{TAIL} = 2 \cdot I_{bias} = 2 \cdot 100 \mu\text{A} = 200 \mu\text{A}. \quad (12)$$

This current splits evenly between the two diff-pair branches, so each transistor's drain current can be calculated as:

$$I_{D1} = I_{D2} = I_{D3} = I_{D4} = \frac{I_{TAIL}}{2} = 100 \mu\text{A}. \quad (13)$$

Using our chosen widths and given our calculated I_D , we could use the LUTs to obtain $g_{m1} = 0.672 \text{ mS}$ and $g_{m4} = 0.489 \text{ mS}$.

For a single-stage OTA loaded by C_L , the gain-bandwidth product can be calculated as follows:

$$\text{GBW} = \frac{g_{m1}}{2\pi C_L} = \frac{0.672 \times 10^{-3}}{2\pi \cdot 0.5 \times 10^{-12}} = 213.9 \text{ MHz}. \quad (14)$$

This satisfies the design requirement of $\text{GBW} \geq 200 \text{ MHz}$.

Assuming a dominant-pole response, the phase margin can be approximated as:

$$\text{PM} \approx 90^\circ - \tan^{-1}(f_u), \quad (15)$$

where $f_u = \text{GBW}$ is the unity-gain frequency. For the 5T OTA, $f_{p2} \approx g_{m3}/(2\pi \cdot 2C_{gs3})$. Calculating PM, we get that $\text{PM} = 88.4$, which satisfies the requirement of $\text{PM} \geq 60^\circ$.

The DC gain of the single-stage 5T OTA is:

$$A_v = g_{m1} \cdot (r_{o,N} \parallel r_{o,P}), \quad (16)$$

where $r_{o,N}$ and $r_{o,P}$ are the small-signal output resistances of M_1 and M_4 respectively, computed from the LUT intrinsic gain values:

$$r_o = \frac{(g_m/g_{ds})}{g_m}. \quad (17)$$

From the LUTs given when $V_{DS} = 0.6 \text{ V}$, we can find that $r_{o,N} = \frac{57.30}{0.672 \times 10^{-3}} = 85.27 \text{ k}\Omega$ and $r_{o,P} = \frac{74.75}{0.489 \times 10^{-3}} = 152.86 \text{ k}\Omega$. Then, the theoretical DC gain can be found as:

$$A_v = g_{m1} \cdot R_{out} = (0.672 \times 10^{-3}) \cdot (54.74 \times 10^3) = 36.78 \text{ V/V}, \quad (18)$$

or in decibels:

$$A_v(\text{dB}) = 20 \log_{10}(36.78) = 31.31 \text{ dB}. \quad (19)$$

2.2 Calculated vs Simulated Performance

The theoretically derived parameters were implemented within Xschem using the Skywater 130nm PDK. The schematic made in Xschem and the simulation results are shown below.

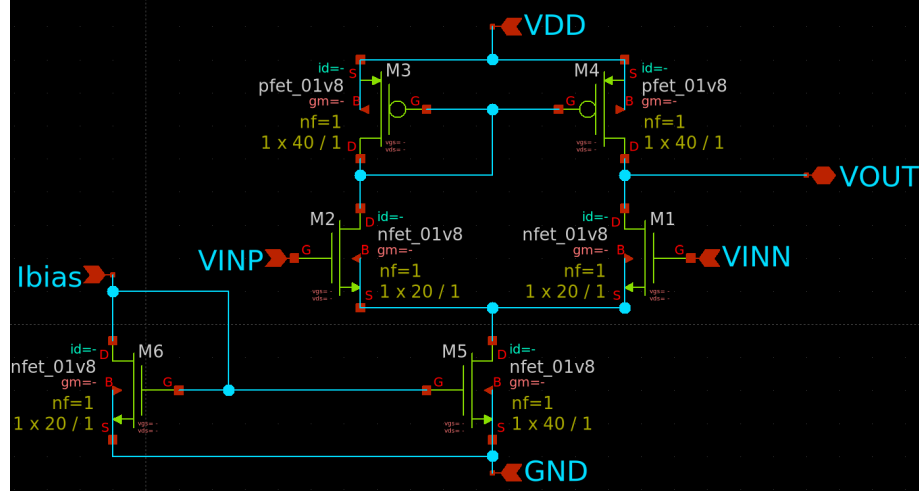


Figure 4: Complete Schematic of high speed 5T OTA

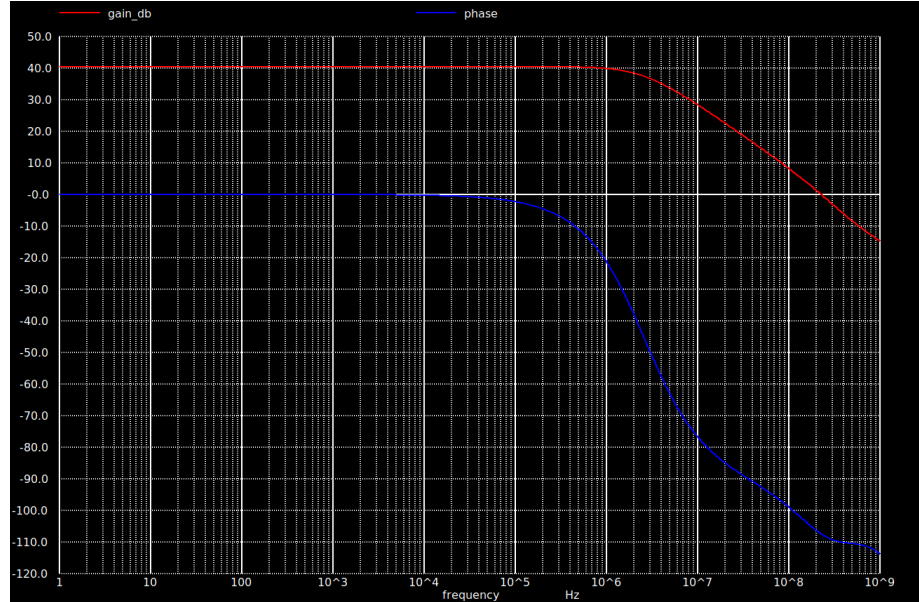


Figure 5: Simulation results of high speed 5T OTA

```

No. of Data Rows : 901
binary raw file "/home/azhang2718/.xschem/simulations/5t_ota_ac.raw"
dc_gain          = 4.050658e+01
fu               = 2.268483e+08
phase            = -1.073758e+02
dc_gain = 4.050658e+01
fu = 2.268483e+08
phase = -1.07376e+02

```

Figure 6: Simulation results of high speed 5T OTA

The hand-calculated DC gain of 31.3 dB significantly underestimates the simulated value of 40.5 dB, a difference of approximately 9 dB. This discrepancy is dominated by the V_{DS} dependence of the small-signal output resistance r_o , which is not captured by the gm/ID LUT. The LUT was generated at a fixed $|V_{DS}| = 0.6$ V, but the actual operating points obtained from the DC simulation are $V_{DS,M2} = 0.388$ V and $|V_{DS,M4}| = 0.904$ V, both of which are significantly different from the LUT bias point.

In short-channel MOSFETs, channel-length modulation (CLM) cause r_o to decrease with increasing V_{DS} . Thus, the NMOS device M_2 , operating at a lower V_{DS} than the LUT, exhibits a higher actual r_o than the LUT predicts. As such, the actual simulated value of the DC gain is shown to be much higher than the simulated version, and actually meets the requirements of the gain.

The simulated GBW of 226.8 MHz is relatively close to the hand-calculated value of 213.9 MHz. The small difference can be attributed to the slightly higher actual g_{m1} in simulation (928.5 μ S) compared to the LUT-derived value (672 μ S used in the hand calculation). However, the value is also balanced by the additional output capacitance from the parasitic C_{gd} of M_2 and M_4 and the drain-bulk junction capacitances $C_{db,M2}$ and $C_{db,M4}$, which add to $C_L = 0.5$ pF. The two effects seem to cancel, giving a simulated GBW close to the hand-calculated prediction.

The hand calculation of the phase margin predicts a phase margin near 88° , while the simulated PM is 72.6° . The discrepancy arises because the single-pole model neglects the non-dominant pole at the gate of the PMOS current mirror, located at:

$$f_{p2} \approx \frac{g_{m3}}{2\pi \cdot 2C_{gs3}}. \quad (20)$$

This pole introduces additional phase lag at frequencies approaching f_u , which then explains why the simulated phase margin is lower than the estimated phase margin. Even though the simulated PM is below the hand-calculated value, the simulated PM still exceeds the 60° specification with a comfortable margin.